

Conditional Probability

ANSWER KEY



Basic probability rules

- 1 A bag contains 5 red marbles, 3 blue marbles, and 2 green marbles. Find the probability of drawing a blue marble.

$$\text{Total marbles} = 5 + 3 + 2 = 10. P(\text{blue}) = \frac{3}{10}.$$

- 2 A fair six-sided die is rolled. Find the probability of rolling an even number or a 5.

Even numbers: 2, 4, 6 (3 outcomes). Adding the outcome 5 (not already counted): total favorable outcomes = 4. $P(\text{even or } 5) = \frac{4}{6} = \frac{2}{3}.$

Two-way tables and conditional probability

- 3 A survey of 300 people found: 90 own a dog and a cat, 60 own only a dog, 40 own only a cat, and 110 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.

$$\text{Total dog owners: } 90 + 60 = 150. \text{ Of those, 90 also own a cat. } P(\text{cat} \mid \text{dog}) = \frac{90}{150} = 0.6, \text{ or } 60\%.$$

- 4 Using the same survey data, find the probability a randomly selected person owns a dog, given that they own a cat.

$$\text{Total cat owners: } 90 + 40 = 130. \text{ Of those, 90 also own a dog. } P(\text{dog} \mid \text{cat}) = \frac{90}{130} = \frac{9}{13} \approx 0.692, \text{ or about } 69.2\%.$$

Independent vs. dependent events

- 5 Explain the difference between independent and dependent events, and classify the following: drawing two cards from a deck without replacement.

Independent events are ones where the outcome of one event does not affect the probability of the other; dependent events are ones where the outcome of one event DOES affect the probability of the other. Drawing two cards without replacement is dependent, because removing the first card changes the total number of cards and the composition of the remaining deck, which affects the probability of the second draw.

- 6 A jar has 4 red and 6 blue chips. Find the probability of drawing two red chips in a row, without replacement.

$$P(\text{first red}) = \frac{4}{10}. \text{ After removing one red chip, } P(\text{second red} \mid \text{first red}) = \frac{3}{9}.$$
$$P(\text{both red}) = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}.$$

The multiplication rule for independent events

- 7 A fair coin is flipped and a fair six-sided die is rolled. Find the probability of getting heads and rolling a 4.

Since the coin flip and die roll are independent: $P(\text{heads and } 4) = P(\text{heads}) \times P(4) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.

Probability with replacement vs. without replacement

- 8 A deck of 10 cards numbered 1-10 has one card drawn, replaced, then a second card drawn. Find the probability both cards drawn are greater than 7.

Cards greater than 7: 8, 9, 10 (3 out of 10). Since the card is replaced, the draws are independent:
 $P(\text{both} > 7) = \frac{3}{10} \times \frac{3}{10} = \frac{9}{100}$.

- 9 Using the same deck of 10 cards but WITHOUT replacement this time, find the probability both cards drawn are greater than 7.

$P(\text{first} > 7) = \frac{3}{10}$. After removing one such card, $P(\text{second} > 7 \mid \text{first} > 7) = \frac{2}{9}$.
 $P(\text{both} > 7) = \frac{3}{10} \times \frac{2}{9} = \frac{6}{90} = \frac{1}{15}$.

Interpreting conditional probability in real-world contexts

- 10 A medical test is 95% accurate for detecting a disease that affects 1% of the population. Explain, in words, why a positive test result does not necessarily mean a 95% chance of having the disease, referencing the low overall disease rate.

Because the disease is rare (only 1% of the population has it), even with a 95%-accurate test, the vast majority of people tested are disease-free, so a meaningful number of false positives (healthy people incorrectly testing positive) will occur alongside the true positives. This means the probability of actually having the disease GIVEN a positive test result (a conditional probability) can be much lower than 95%, since it depends on the relative sizes of the true-positive and false-positive groups, not just the test's accuracy rate alone.

The complement rule

- 11 A spinner has 8 equal sections numbered 1-8. Find the probability of NOT landing on a number greater than 6, using the complement rule.

$P(\text{greater than } 6) = P(7 \text{ or } 8) = \frac{2}{8} = \frac{1}{4}$. By the complement rule: $P(\text{not greater than } 6) = 1 - \frac{1}{4} = \frac{3}{4}$.

Probability trees for multi-stage events

- 12** A box has 3 red and 2 white balls. Two balls are drawn without replacement. Find the probability that the first is red and the second is white.

$$P(\text{first red}) = \frac{3}{5}. \text{ After removing one red ball, } P(\text{second white} \mid \text{first red}) = \frac{2}{4} = \frac{1}{2}.$$

$$P(\text{red then white}) = \frac{3}{5} \times \frac{1}{2} = \frac{3}{10}.$$

- 13** Using the same box (3 red, 2 white, no replacement), find the probability of drawing two balls of the same color.

$$P(\text{both red}) = \frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = \frac{3}{10}. P(\text{both white}) = \frac{2}{5} \times \frac{1}{4} = \frac{2}{20} = \frac{1}{10}.$$

$$P(\text{same color}) = \frac{3}{10} + \frac{1}{10} = \frac{4}{10} = \frac{2}{5}.$$

Comparing theoretical and experimental probability

- 14** A fair coin is flipped 50 times and lands heads 32 times. Compare the experimental probability of heads to the theoretical probability, and explain a possible reason for the difference.

Experimental probability: $\frac{32}{50} = 0.64$, or 64%. Theoretical probability: 0.5, or 50%. The difference likely arises from natural random variation in a finite number of trials — with only 50 flips, results can deviate noticeably from the theoretical probability by chance; as the number of trials increases, the experimental probability tends to converge toward the theoretical probability (the law of large numbers).

Combining conditional probability with two-way tables

- 15** A two-way table of 400 students shows: 180 play an instrument and are in band; 20 play an instrument but are not in band; 40 don't play an instrument but are in band; 160 neither play an instrument nor are in band. Find the probability a student is in band, given that they play an instrument.

Total who play an instrument: $180 + 20 = 200$. Of those, 180 are in band.

$$P(\text{band} \mid \text{plays instrument}) = \frac{180}{200} = 0.9, \text{ or } 90\%.$$

- 16** Using the same table (180 instrument+band, 20 instrument only, 40 band only, 160 neither, total 400), find the probability that a randomly selected student plays an instrument OR is in band (not both counted twice).

Using the addition rule: $P(\text{instrument or band}) = P(\text{instrument}) + P(\text{band}) - P(\text{both})$.

$$P(\text{instrument}) = \frac{200}{400} = 0.5. P(\text{band}) = \frac{180 + 40}{400} = \frac{220}{400} = 0.55. P(\text{both}) = \frac{180}{400} = 0.45.$$

$$P(\text{instrument or band}) = 0.5 + 0.55 - 0.45 = 0.6, \text{ or } 60\%.$$

A multi-part conditional probability problem

- 17 This question has three parts. A school has 600 students: 350 take Spanish, 200 take French, and 50 take both languages. (a) Find the probability a randomly selected student takes at least one of the two languages. (b) Find the probability a student takes Spanish, given that they take at least one language. (c) Find the probability a student takes neither language.

(a) Using the addition rule:

$$P(\text{Spanish or French}) = P(\text{Spanish}) + P(\text{French}) - P(\text{both}) = \frac{350}{600} + \frac{200}{600} - \frac{50}{600} = \frac{500}{600} = \frac{5}{6}.$$

Students taking at least one language: 500. Students taking Spanish (including both): 350.

$$P(\text{Spanish} \mid \text{at least one}) = \frac{350}{500} = 0.7, \text{ or } 70\%. \quad (c) \text{ Students taking neither: } 600 - 500 = 100.$$

$$P(\text{neither}) = \frac{100}{600} = \frac{1}{6}.$$

Multiple choice — easy

- 18 A bag contains 9 red marbles, 12 blue marbles, and 4 green marbles. Find the probability of drawing a blue marble.

a. $\frac{13}{25}$

b. $\frac{12}{25}$ ✓

c. $\frac{9}{25}$

d. $\frac{4}{25}$

Total marbles = $9 + 12 + 4 = 25$. $P(\text{blue}) = \frac{12}{25} = \frac{12}{25}$.

- 19 A bag contains 11 orange marbles, 6 purple marbles, and 6 silver marbles. Find the probability of drawing a silver marble.

a. $\frac{7}{23}$

b. $\frac{11}{23}$

c. $\frac{5}{23}$

d. $\frac{6}{23}$ ✓

Total marbles = $11 + 6 + 6 = 23$. $P(\text{silver}) = \frac{6}{23} = \frac{6}{23}$.

- 20 A bag contains 8 orange marbles, 11 purple marbles, and 9 silver marbles. Find the probability of drawing a purple marble.

a. $\frac{11}{28}$ ✓

b. $\frac{3}{7}$

c. $\frac{9}{28}$

d. $\frac{2}{7}$

Total marbles = $8 + 11 + 9 = 28$. $P(\text{purple}) = \frac{11}{28} = \frac{11}{28}$.

- 21 A bag contains 4 red marbles, 9 blue marbles, and 6 green marbles. Find the probability of drawing a blue marble.

a. $\frac{9}{19}$ ✓

b. $\frac{6}{19}$

c. $\frac{4}{19}$

d. $\frac{10}{19}$

Total marbles = $4 + 9 + 6 = 19$. $P(\text{blue}) = \frac{9}{19} = \frac{9}{19}$.

- 22 A bag contains 4 black marbles, 4 white marbles, and 9 yellow marbles. Find the probability of drawing a black marble.

a. $\frac{4}{17}$ ✓

b. $\frac{5}{17}$

c. $\frac{9}{17}$

d. $\frac{3}{17}$

Total marbles = $4 + 4 + 9 = 17$. $P(\text{black}) = \frac{4}{17} = \frac{4}{17}$.

- 23** A bag contains 8 black marbles, 2 white marbles, and 10 yellow marbles. Find the probability of drawing a black marble.

a. $\frac{2}{5}$ ✓

b. $\frac{1}{2}$

c. $\frac{1}{10}$

d. $\frac{9}{20}$

Total marbles = $8 + 2 + 10 = 20$. $P(\text{black}) = \frac{8}{20} = \frac{2}{5}$.

- 24** A bag contains 9 red marbles, 10 blue marbles, and 12 green marbles. Find the probability of drawing a red marble.

a. $\frac{12}{31}$

b. $\frac{10}{31}$

c. $\frac{9}{31}$ ✓

d. $\frac{8}{31}$

Total marbles = $9 + 10 + 12 = 31$. $P(\text{red}) = \frac{9}{31} = \frac{9}{31}$.

- 25** A bag contains 5 red marbles, 2 blue marbles, and 8 green marbles. Find the probability of drawing a green marble.

a. $\frac{2}{15}$

b. $\frac{3}{5}$

c. $\frac{1}{3}$

d. $\frac{8}{15}$ ✓

Total marbles = $5 + 2 + 8 = 15$. $P(\text{green}) = \frac{8}{15} = \frac{8}{15}$.

- 26** A bag contains 6 black marbles, 8 white marbles, and 5 yellow marbles. Find the probability of drawing a black marble.

- a. $\frac{7}{19}$
- b. $\frac{8}{19}$
- c. $\frac{6}{19}$ ✓
- d. $\frac{5}{19}$

$$\text{Total marbles} = 6 + 8 + 5 = 19. P(\text{black}) = \frac{6}{19} = \frac{6}{19}.$$

- 27** A bag contains 5 orange marbles, 10 purple marbles, and 8 silver marbles. Find the probability of drawing a purple marble.

- a. $\frac{8}{23}$
- b. $\frac{5}{23}$
- c. $\frac{10}{23}$ ✓
- d. $\frac{11}{23}$

$$\text{Total marbles} = 5 + 10 + 8 = 23. P(\text{purple}) = \frac{10}{23} = \frac{10}{23}.$$

- 28** A bag contains 5 black marbles, 9 white marbles, and 5 yellow marbles. Find the probability of drawing a white marble.

- a. $\frac{5}{19}$
- b. $\frac{10}{19}$
- c. $\frac{9}{19}$ ✓
- d. $\frac{8}{19}$

$$\text{Total marbles} = 5 + 9 + 5 = 19. P(\text{white}) = \frac{9}{19} = \frac{9}{19}.$$

- 29 A bag contains 7 black marbles, 8 white marbles, and 5 yellow marbles. Find the probability of drawing a black marble.

- a. $\frac{3}{10}$
- b. $\frac{2}{5}$
- c. $\frac{1}{4}$
- d. $\frac{7}{20}$ ✓

Total marbles = $7 + 8 + 5 = 20$. $P(\text{black}) = \frac{7}{20} = \frac{7}{20}$.

- 30 A bag contains 7 black marbles, 12 white marbles, and 9 yellow marbles. Find the probability of drawing a yellow marble.

- a. $\frac{9}{28}$ ✓
- b. $\frac{1}{4}$
- c. $\frac{5}{14}$
- d. $\frac{3}{7}$

Total marbles = $7 + 12 + 9 = 28$. $P(\text{yellow}) = \frac{9}{28} = \frac{9}{28}$.

- 31 A bag contains 8 orange marbles, 11 purple marbles, and 7 silver marbles. Find the probability of drawing a purple marble.

- a. $\frac{6}{13}$
- b. $\frac{4}{13}$
- c. $\frac{11}{26}$ ✓
- d. $\frac{7}{26}$

Total marbles = $8 + 11 + 7 = 26$. $P(\text{purple}) = \frac{11}{26} = \frac{11}{26}$.

- 32 A bag contains 5 orange marbles, 9 purple marbles, and 10 silver marbles. Find the probability of drawing a purple marble.

a. $\frac{3}{8}$ ✓

b. $\frac{1}{3}$

c. $\frac{5}{12}$

d. $\frac{5}{24}$

Total marbles = $5 + 9 + 10 = 24$. $P(\text{purple}) = \frac{9}{24} = \frac{3}{8}$.

Multiple choice — medium

- 33 A survey of 83 people found: 47 own a car and a bicycle, 5 own only a car, 5 own only a bicycle, and 26 own neither. Find the probability a randomly selected person owns a bicycle, given that they own a car.

a. $\frac{47}{83}$

b. 1

c. $\frac{25}{26}$

d. $\frac{47}{52}$ ✓

Total who own a car: $47 + 5 = 52$. Of those, 47 also own a bicycle. $P = \frac{47}{52} = \frac{47}{52}$.

- 34 A survey of 73 people found: 21 own a dog and a cat, 9 own only a dog, 20 own only a cat, and 23 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.

a. $\frac{21}{41}$

b. $\frac{21}{73}$

c. $\frac{7}{10}$ ✓

d. $\frac{41}{30}$

Total who own a dog: $21 + 9 = 30$. Of those, 21 also own a cat. $P = \frac{21}{30} = \frac{7}{10}$.

- 35 A survey of 83 people found: 21 own a car and a bicycle, 16 own only a car, 26 own only a bicycle, and 20 own neither. Find the probability a randomly selected person owns a bicycle, given that they own a car.

- a. $\frac{21}{83}$
- b. $\frac{21}{47}$
- c. $\frac{21}{37}$ ✓
- d. $\frac{47}{37}$

Total who own a car: $21 + 16 = 37$. Of those, 21 also own a bicycle. $P = \frac{21}{37} = \frac{21}{37}$.

- 36 A survey of 71 people found: 27 own a piano and a guitar, 30 own only a piano, 8 own only a guitar, and 6 own neither. Find the probability a randomly selected person owns a guitar, given that they own a piano.

- a. $\frac{27}{71}$
- b. $\frac{9}{19}$ ✓
- c. $\frac{27}{35}$
- d. $\frac{35}{57}$

Total who own a piano: $27 + 30 = 57$. Of those, 27 also own a guitar. $P = \frac{27}{57} = \frac{9}{19}$.

- 37 A survey of 98 people found: 46 own a smartphone and a smartwatch, 11 own only a smartphone, 26 own only a smartwatch, and 15 own neither. Find the probability a randomly selected person owns a smartwatch, given that they own a smartphone.

- a. $\frac{24}{19}$
- b. $\frac{23}{36}$
- c. $\frac{23}{49}$
- d. $\frac{46}{57}$ ✓

Total who own a smartphone: $46 + 11 = 57$. Of those, 46 also own a smartwatch. $P = \frac{46}{57} = \frac{46}{57}$.

- 38 A survey of 134 people found: 32 own a piano and a guitar, 35 own only a piano, 36 own only a guitar, and 31 own neither. Find the probability a randomly selected person owns a guitar, given that they own a piano.

- a. $\frac{8}{17}$
- b. $\frac{68}{67}$
- c. $\frac{16}{67}$
- d. $\frac{32}{67}$ ✓

Total who own a piano: $32 + 35 = 67$. Of those, 32 also own a guitar. $P = \frac{32}{67} = \frac{32}{67}$.

- 39 A survey of 92 people found: 22 own a smartphone and a smartwatch, 14 own only a smartphone, 22 own only a smartwatch, and 34 own neither. Find the probability a randomly selected person owns a smartwatch, given that they own a smartphone.

- a. $\frac{1}{2}$
- b. $\frac{11}{9}$
- c. $\frac{11}{18}$ ✓
- d. $\frac{11}{46}$

Total who own a smartphone: $22 + 14 = 36$. Of those, 22 also own a smartwatch. $P = \frac{22}{36} = \frac{11}{18}$.

- 40 A survey of 80 people found: 20 own a car and a bicycle, 29 own only a car, 13 own only a bicycle, and 18 own neither. Find the probability a randomly selected person owns a bicycle, given that they own a car.

- a. $\frac{33}{49}$
- b. $\frac{20}{49}$ ✓
- c. $\frac{1}{4}$
- d. $\frac{20}{33}$

Total who own a car: $20 + 29 = 49$. Of those, 20 also own a bicycle. $P = \frac{20}{49} = \frac{20}{49}$.

- 41 A survey of 78 people found: 43 own a dog and a cat, 23 own only a dog, 7 own only a cat, and 5 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.

a. $\frac{43}{66}$ ✓

b. $\frac{43}{78}$

c. $\frac{43}{50}$

d. $\frac{25}{33}$

Total who own a dog: $43 + 23 = 66$. Of those, 43 also own a cat. $P = \frac{43}{66} = \frac{43}{66}$.

- 42 A survey of 53 people found: 15 own a dog and a cat, 22 own only a dog, 6 own only a cat, and 10 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.

a. $\frac{15}{53}$

b. $\frac{5}{7}$

c. $\frac{15}{37}$ ✓

d. $\frac{21}{37}$

Total who own a dog: $15 + 22 = 37$. Of those, 15 also own a cat. $P = \frac{15}{37} = \frac{15}{37}$.

- 43 A survey of 97 people found: 59 own a car and a bicycle, 13 own only a car, 12 own only a bicycle, and 13 own neither. Find the probability a randomly selected person owns a bicycle, given that they own a car.

a. $\frac{71}{72}$

b. $\frac{59}{71}$

c. $\frac{59}{97}$

d. $\frac{59}{72}$ ✓

Total who own a car: $59 + 13 = 72$. Of those, 59 also own a bicycle. $P = \frac{59}{72} = \frac{59}{72}$.

- 44 A survey of 75 people found: 20 own a dog and a cat, 22 own only a dog, 7 own only a cat, and 26 own neither. Find the probability a randomly selected person owns a cat, given that they own a dog.

- a. $\frac{9}{14}$
- b. $\frac{20}{27}$
- c. $\frac{10}{21}$ ✓
- d. $\frac{4}{15}$

Total who own a dog: $20 + 22 = 42$. Of those, 20 also own a cat. $P = \frac{20}{42} = \frac{10}{21}$.

- 45 A survey of 67 people found: 15 own a piano and a guitar, 36 own only a piano, 6 own only a guitar, and 10 own neither. Find the probability a randomly selected person owns a guitar, given that they own a piano.

- a. $\frac{5}{7}$
- b. $\frac{7}{17}$
- c. $\frac{5}{17}$ ✓
- d. $\frac{15}{67}$

Total who own a piano: $15 + 36 = 51$. Of those, 15 also own a guitar. $P = \frac{15}{51} = \frac{5}{17}$.

- 46 A survey of 107 people found: 35 own a smartphone and a smartwatch, 23 own only a smartphone, 33 own only a smartwatch, and 16 own neither. Find the probability a randomly selected person owns a smartwatch, given that they own a smartphone.

- a. $\frac{35}{58}$ ✓
- b. $\frac{35}{68}$
- c. $\frac{34}{29}$
- d. $\frac{35}{107}$

Total who own a smartphone: $35 + 23 = 58$. Of those, 35 also own a smartwatch. $P = \frac{35}{58} = \frac{35}{58}$.

- 47 A survey of 84 people found: 20 own a smartphone and a smartwatch, 8 own only a smartphone, 19 own only a smartwatch, and 37 own neither. Find the probability a randomly selected person owns a smartwatch, given that they own a smartphone.

a. $\frac{39}{28}$

b. $\frac{5}{7}$ ✓

c. $\frac{5}{21}$

d. $\frac{20}{39}$

Total who own a smartphone: $20 + 8 = 28$. Of those, 20 also own a smartwatch. $P = \frac{20}{28} = \frac{5}{7}$.

Multiple choice — hard

- 48 Two independent events occur: a spinner (5 equal sections) lands on section 1, and a fair six-sided die shows an even number. Find the probability both events occur.

a. $\frac{1}{10}$ ✓

b. $\frac{1}{5}$

c. $\frac{1}{20}$

d. $\frac{7}{10}$

Since the events are independent: $P(\text{both}) = P_1 \times P_2 = \frac{1}{5} \times \frac{1}{2} = \frac{1}{10}$.

- 49 Two independent events occur: a spinner (5 equal sections) lands on section 1, and a card drawn from a standard deck is a heart. Find the probability both events occur.

a. $\frac{1}{20}$ ✓

b. $\frac{3}{20}$

c. $\frac{1}{10}$

d. $\frac{9}{20}$

Since the events are independent: $P(\text{both}) = P_1 \times P_2 = \frac{1}{5} \times \frac{1}{4} = \frac{1}{20}$.

- 50 Two independent events occur: a card drawn from a standard deck is a heart, and a fair coin lands heads. Find the probability both events occur.

- a. $\frac{1}{4}$
- b. $\frac{9}{40}$
- c. $\frac{1}{8}$ ✓
- d. $\frac{3}{4}$

Since the events are independent: $P(\text{both}) = P_1 \times P_2 = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$.

- 51 Two independent events occur: a fair six-sided die shows an even number, and a spinner (5 equal sections) lands on section 1. Find the probability both events occur.

- a. $\frac{3}{20}$
- b. $\frac{7}{10}$
- c. $\frac{1}{10}$ ✓
- d. $\frac{1}{5}$

Since the events are independent: $P(\text{both}) = P_1 \times P_2 = \frac{1}{2} \times \frac{1}{5} = \frac{1}{10}$.

- 52 Two independent events occur: a fair coin lands heads, and a fair six-sided die shows an even number. Find the probability both events occur.

- a. $\frac{1}{2}$
- b. 1
- c. $\frac{7}{20}$
- d. $\frac{1}{4}$ ✓

Since the events are independent: $P(\text{both}) = P_1 \times P_2 = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

- 53 A box has 5 red and 4 white balls. Two balls are drawn without replacement. Find the probability that the first is red and the second is white.

a. $\frac{5}{18}$ ✓

b. $\frac{17}{45}$

c. $\frac{5}{9}$

d. $\frac{59}{180}$

$P(\text{first}) = \frac{5}{9}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{1}{2}$. $P(\text{both}) = \frac{5}{9} \times \frac{1}{2} = \frac{5}{18}$.

- 54 A box has 4 red and 7 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.

a. $\frac{28}{55}$

b. $\frac{7}{11}$

c. $\frac{67}{220}$

d. $\frac{14}{55}$ ✓

$P(\text{first}) = \frac{7}{11}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{2}{5}$. $P(\text{both}) = \frac{7}{11} \times \frac{2}{5} = \frac{14}{55}$.

- 55 A box has 6 red and 6 white balls. Two balls are drawn without replacement. Find the probability that the first is red and the second is white.

a. $\frac{6}{11}$

b. $\frac{3}{11}$ ✓

c. $\frac{71}{220}$

d. $\frac{1}{2}$

$P(\text{first}) = \frac{1}{2}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{6}{11}$. $P(\text{both}) = \frac{1}{2} \times \frac{6}{11} = \frac{3}{11}$.

- 56 A box has 6 red and 3 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.

- a. $\frac{1}{3}$
- b. $\frac{1}{4}$ ✓
- c. $\frac{3}{10}$
- d. $\frac{1}{2}$

$P(\text{first}) = \frac{1}{3}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{3}{4}$. $P(\text{both}) = \frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$.

- 57 A box has 8 red and 7 white balls. Two balls are drawn without replacement. Find the probability that the first is white and the second is red.

- a. $\frac{7}{15}$
- b. $\frac{19}{60}$
- c. $\frac{4}{15}$ ✓
- d. $\frac{8}{15}$

$P(\text{first}) = \frac{7}{15}$. After removing one ball, $P(\text{second} \mid \text{first}) = \frac{4}{7}$. $P(\text{both}) = \frac{7}{15} \times \frac{4}{7} = \frac{4}{15}$.

- 58 A spinner has 15 equal sections numbered 1 to 15. Find the probability of NOT landing on one of the 5 sections numbered 1 through 5, using the complement rule.

- a. $\frac{2}{5}$
- b. $\frac{4}{15}$
- c. $\frac{1}{3}$
- d. $\frac{2}{3}$ ✓

The probability of landing on sections 1 through 5 is $\frac{5}{15}$. By the complement rule: $P(\text{not}) = 1 - \frac{5}{15} = \frac{2}{3}$.

- 59 A spinner has 15 equal sections numbered 1 to 15. Find the probability of NOT landing on one of the 2 sections numbered 1 through 2, using the complement rule.

a. $\frac{13}{15}$ ✓

b. $\frac{2}{15}$

c. $\frac{1}{5}$

d. $\frac{1}{15}$

The probability of landing on sections 1 through 2 is $\frac{2}{15}$. By the complement rule: $P(not) = 1 - \frac{2}{15} = \frac{13}{15}$.

- 60 A spinner has 7 equal sections numbered 1 to 7. Find the probability of NOT landing on one of the 2 sections numbered 1 through 2, using the complement rule.

a. $\frac{3}{7}$

b. $\frac{5}{7}$ ✓

c. $\frac{1}{7}$

d. $\frac{2}{7}$

The probability of landing on sections 1 through 2 is $\frac{2}{7}$. By the complement rule: $P(not) = 1 - \frac{2}{7} = \frac{5}{7}$.

- 61 A spinner has 9 equal sections numbered 1 to 9. Find the probability of NOT landing on one of the 8 sections numbered 1 through 8, using the complement rule.

a. $\frac{1}{3}$

b. $\frac{1}{9}$ ✓

c. $\frac{8}{9}$

d. $\frac{7}{9}$

The probability of landing on sections 1 through 8 is $\frac{8}{9}$. By the complement rule: $P(not) = 1 - \frac{8}{9} = \frac{1}{9}$.

- 62 A spinner has 13 equal sections numbered 1 to 13. Find the probability of NOT landing on one of the 10 sections numbered 1 through 10, using the complement rule.

a. $\frac{10}{13}$

b. $\frac{3}{13}$ ✓

c. $\frac{9}{13}$

d. $\frac{11}{13}$

The probability of landing on sections 1 through 10 is $\frac{10}{13}$. By the complement rule: $P(not) = 1 - \frac{10}{13} = \frac{3}{13}$.